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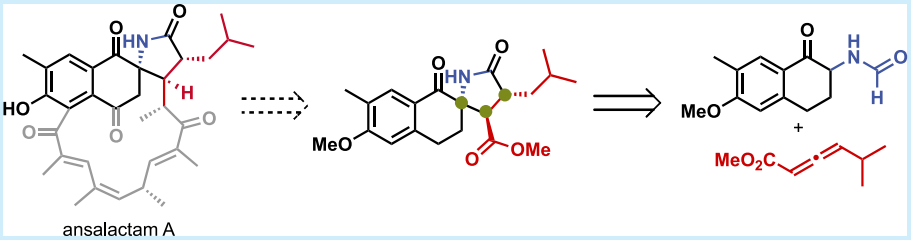
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Stereoselective Synthesis of the Spirocyclic γ-Lactam Core of the Ansalactams

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ABSTRACT: Ansalactam A is an ansa macrolide natural product that contains a densely functionalized spiro-γ-lactam core containing three contiguous stereocenters. This unusual motif distinguishes it from other members of this family and represents a significant synthetic challenge. Herein, we report the development of a stereoselective formal [3+2] cycloaddition reaction for the construction of this key spiro-γ-lactam motif for the first time, thereby enabling access to the northern domain of ansalactam A.

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Ansalactams A–D represent a subclass of ansamycin polyketides that were isolated by Moore and Fenical from *Streptomyces* sp. strain CNH-189 isolated from marine sediments.[1,2] These natural products are members of the broader ansamycin family that includes biologically potent molecules such as rifamycin SV and geldanamycin.[3,4] In addition to their biological activity, the ansamycins have complex biosynthetic origins and are collectively formed from a common aromatic acid 3-amino-5-hydroxybenzoate precursor.[1] The ansamycins are structurally differentiated by variations in a shared spiro-γ-lactam dihydronaphthoquinone core, further bridged by a polyketide chain forming a 16-membered macrocyclic ring [**1**–**4** (Figure 1)].

As part of our broader program targeting bioactive marine natural products,[5] we became interested in developing a synthetic route to the spiro-γ-lactam system unique to the ansalactams. This isobutyryl-substituted lactam has been hypothesized by Trauner to arise biosynthetically through a 5-*exo* or 5-*endo* radical cyclization. However, preliminary synthetic studies that aimed to realize this biomimetic logic have been unsuccessful to date (Figure 2a).[6] We envisioned a [3+2] cycloaddition approach to preparing the spiro-γ-lactam core, leveraging the well-studied steric tolerance and stereocontrol of these reactions (Figure 2b).[7,8] While this approach was attractive, it was not clear at the outset of this work what the identity of the 1,3-dipole (represented by **8**) or dipolarophile (represented by **9**) would be, as the proposed disconnections required inversion of the native reactivity of an oxy-azomethine ylide precursor such as **8**. Attempted formation of **8** from a formamide precursor such as **10** may result in the formation of an undesired α-isocyanoacetate

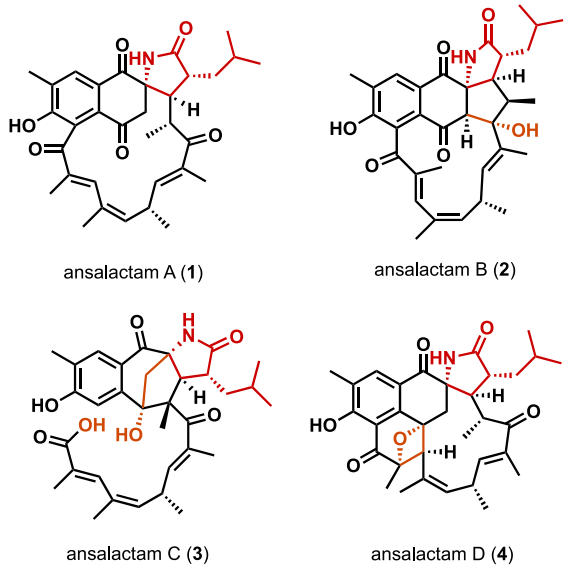


Figure 1. Structures of the ansalactam natural products (**1**–**4**).

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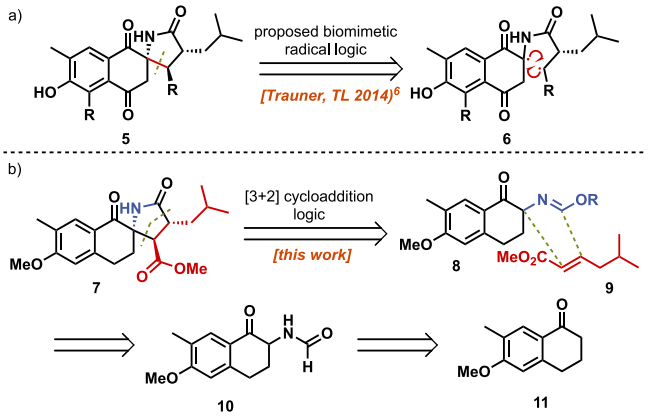
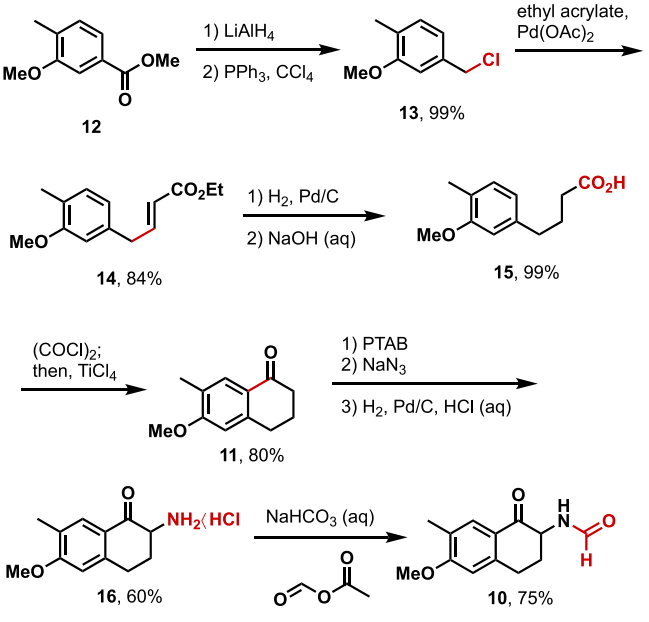


Figure 2. (a) Biomimetic logic presented by Trauner. (b) Retrosynthetic approach to the spirolactam core enabled by a key [3+2] dipolar cycloaddition.

derivative, precluding formation of the targeted spirocycle due to competitive [4+1] cycloaddition reactions that commonly provide an array of heterocycles from these motifs.[9]

To start to explore the reactivity of these systems, we sought to prepare formamide **10** (Scheme 1). Beginning with methyl

Scheme 1. Synthesis of Formamide 10 (PTAB = phenyltrimethylammonium tribromide)



ester **12**, reduction and Appel chlorination afforded benzylic chloride **13** in 99% yield. Subsequent Heck coupling,[10] followed by hydrogenation and saponification, allowed for the preparation of carboxylic acid **15** in excellent yield over three steps. Friedel–Crafts acylation proceeded smoothly to generate ketone **11**.[11] Sequential bromination, nucleophilic azide installation, and hydrogenation afforded the desired 2-aminotetralone **16** in 60% yield. Lastly, N-formylation of **16** provided the targeted formamide **10** in 75% yield. The sequence described here proved to be highly scalable and has allowed for scalable preparation of **10** such that we envision further streamlining and leveraging for gram-scale quantities in the future.

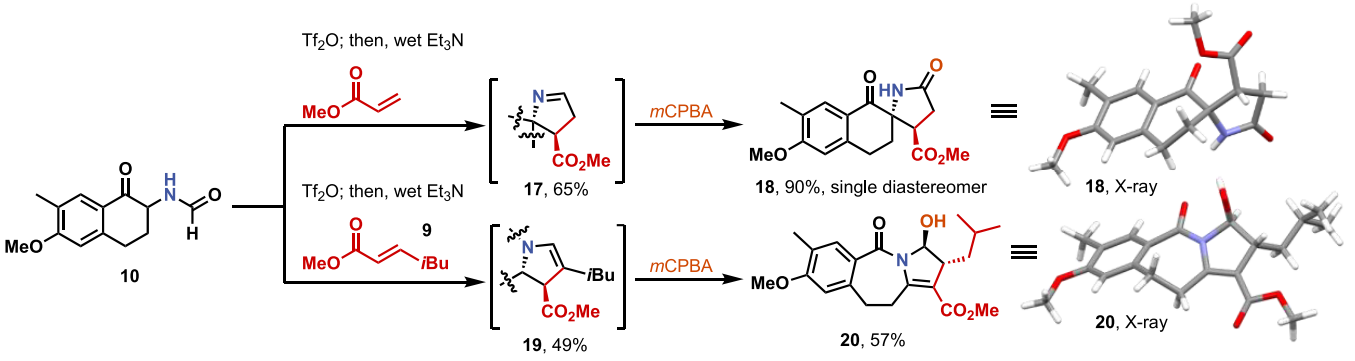
With **10** in hand, we envisioned activation of the formamide group with triflic anhydride, triggering formation of either an oxy-azomethine-type 1,3-dipole or isocyanide. Subsequent reaction with an α,β-unsaturated carbonyl compound, in a concerted or stepwise fashion, would then forge the desired spiro-γ-lactam through a formal [3+2] process. To explore this approach, we first employed methyl acrylate as the reaction partner in the presence of Et_3N (not rigorously dried), resulting in the formation of a five-membered spirocyclic product **17** that possessed the desired regiochemical and stereochemical outcome yet had lost the oxygen atom from the formamide group (Scheme 2). This result suggested that the reaction was proceeding through an isocyanide intermediate, but questions regarding mechanistic aspects of this process remained. Given the stoichiometric quantities of acid present, it is likely that the added amine base is functioning as only an addition proton shuttle and a source of trace water (addition of small quantities of water alone was not as effective).

The oxidation of imine to the corresponding lactam **18** using *m*CBPA provided the target spiro-γ-lactam in 90% yield and as a single isolated diastereomer, the absolute stereochemistry of which was assigned though X-ray analysis. Unfortunately, all attempts to alkylate or otherwise functionalize this compound to install the isobutyl group failed. We then sought to incorporate this sterically demanding group into the dipolarophile component of the key [3+2] reaction. To probe this approach, formamide **10** was activated with triflic anhydride and reacted with enoate **9** (Scheme 2), ultimately providing **20** in a modest 28% yield over the two steps after oxidation of **19** with *m*CPBA. Only one regioisomer was isolated from this reaction, albeit the low yield makes the overall selectivity uncertain. Characterization of the product revealed that a skeletal rearrangement had occurred, and this was further corroborated by X-ray analysis (Scheme 2). At this stage, it is not clear what drives this ring expansion, but we hypothesize that the increased sterics of this system slows the desired addition, allowing alternate reaction pathways to dominate. While this transformation was not relevant to the synthesis of this family of natural products, the structural complexity generated in this process may be of interest in other contexts.

It was clear that a [3+2] cycloaddition could be utilized to forge the desired spirolactam, but further optimization was required to incorporate the isobutyl side chain contained in the natural product. Mechanistically, we originally envisioned this transformation proceeding via azomethine ylide-type reactivity (Figure 3a), but observation of deoxygenated imine product **17** suggests the formation of such an intermediate was unlikely in the presence of strongly electrophilic activating agents (Tf_2O). Instead, we propose formation of a protonated isocyanide intermediate (**22**) under the strongly acidic reaction conditions (Figure 3b). Activation of **10** in the absence of an electrophilic reaction partner yielded oxazole **21** (confirmed by X-ray analysis), further suggesting the intermediacy of such an isocyanide species. Furthermore, this isolated compound is a competent reactant in the desired [3+2] reaction under the same conditions, supporting its formation during the course of the reaction. Control experiments have confirmed the requirement of a strong acid, with no reaction observed under neutral or basic reaction conditions. We propose that protonated isocyanide **21** is critical for driving the observed reactivity and regiochemical outcome as the compound can react as an extended enolate, enabling closure of the ester enol

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Scheme 2. Initial Exploration of the Proposed [3+2] Cycloaddition



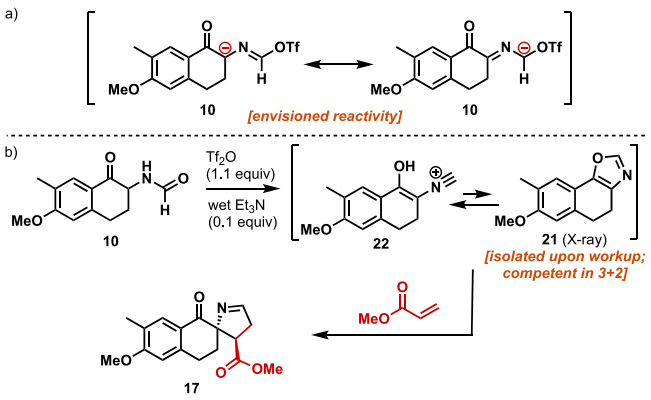
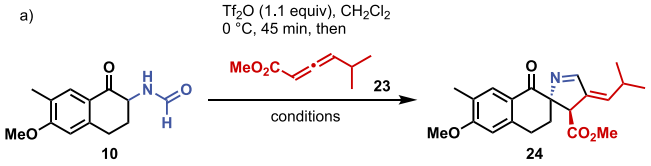


Figure 3. (a) Initially envisioned reactivity. (b) Proposed reaction pathway to provide **17**.



entry	conditions[a]	concentration	24 (%)[b,c]
1	Et_3N (0.1 equiv), **23** (2 equiv) CH_2Cl_2, rt, 16 h	0.1 M	15%
2	Et_3N (0.1 equiv), **23** (2 equiv) toluene, rt, 16 h	0.1 M	trace
3	Et_3N (0.1 equiv), **23** (2 equiv) CH_2Cl_2, rt, 16 h	1 M	26%
4	Et_3N (0.1 equiv), **23** (2 equiv) CH_2Cl_2, rt, 16 h	4 M	40%
5	Et_3N (0.1 equiv), **23** (2 equiv) CH_2Cl_2, rt, 48 h	4 M	62%
6	Et_3N (1 equiv), **23** (2 equiv) CH_2Cl_2, rt, 16 h	0.1 M	trace

[a]*wet Et_3N used in all cases* [b]*E/Z isomers isolated after column chromatography* [c]*isolated yield of E/Z isomers.*

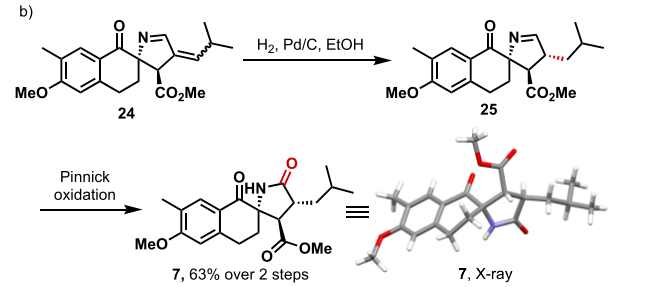


Figure 4. (a) Optimization of the reaction with allene **23**. (b) Conversion of **24** to the northern fragment of ansalactam A (**7**).

in a stepwise fashion; however, we cannot rule out the possibility of the protonated oxazole itself as an intermediate in the reaction. Additional mechanistic studies are underway to further probe this reaction and identify pathways to render it asymmetric.[9]

Given our lack of success utilizing substituted electrophilic alkenes as dipolarophiles, we sought to explore allenes as alternate reactants. It was hypothesized that the planar nature of these compounds would limit the impact that sterics may have played in our preliminary studies of the [3+2] cycloaddition. Our first effort in this regard employed allene **23** under conditions identical to those utilized previously for the alkene reactants (entry 1, Figure 4a). This reaction proceeded in low conversion after 16 h but provided the desired extended imine product **24** in 15% isolated yield as a mixture of *E* and *Z* alkene isomers. We isolate only one diastereomer from these reactions, likely due to the proposed intramolecular sprio center-forming reaction. Further optimization identified dichloromethane as a more suitable solvent at increased concentrations and reaction times, ultimately providing a 62% yield (entries 2−5, Figure 4a). As described above, neutralizing the reaction mixture by employing stoichiometric quantities of triethylamine inhibited the reaction, highlighting the importance of Bronsted acid in this transformation (entry 6, Figure 4a).

With compound **24** in hand, we explored various sequences to reduce the exocyclic alkene and oxidize the imine to reveal the targeted spiro-γ-lactam **7** (Figure 4b). Hydrogenation of **24** in ethanol using Pd/C chemoselectively reduced the extended imine, forming isobutyl-substituted imine **25**. Oxidation of this compound under Pinnick oxidation conditions resulted in the formation of lactam **7** in 63% yield over the two steps as a single isolated diastereomer.[12] Previously employed *m*CPBA conditions did not provide productive yields of this more sterically hindered substrate. Compound **7** represents the entire carbon skeleton of the northern domain of ansalactam A (**1**), lacking only the hemiquinone carbonyl of the natural product. Efforts to accomplish this last oxidation and install the southern polyketide fragment are well underway.

In conclusion, we have developed a stereoselective approach to the core of the ansalactam natural products. Additionally, the methods presented herein are relevant for the synthesis of other natural product scaffolds and bioactive molecules. Efforts to further explore the mechanism and scope of the novel spirocyclic dipolar cycloaddition, to optimize the preparation of the starting materials, and to complete the synthesis of ansalactam A are underway and will be reported in due course.

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■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.1c03782.

Experimental procedures, characterizations of new compounds, X-ray data, and NMR spectra (PDF)

Accession Codes

CCDC 2120591−2120594 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

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■ REFERENCES

(1) Wilson, M. C.; Nam, S.-J.; Gulder, T. A. M.; Kauffman, C. A.; Jensen, P. R.; Fenical, W.; Moore, B. S. Structure and Biosynthesis of the Marine Streptomycete Ansamycin Ansalactam A and Its Distinctive Branched Chain Polyketide Extender Unit. *J. Am. Chem. Soc.* **2011**, *133* (6), 1971−1977.

(2) Le, T. C.; Yang, I.; Yoon, Y. J.; Nam, S.-J.; Fenical, W. Ansalactams B−D Illustrate Further Biosynthetic Plasticity within the Ansamycin Pathway. *Org. Lett.* **2016**, *18* (9), 2256−2259.

(3) Floss, H. G.; Yu, T.-W. Rifamycin - Mode of Action, Resistance, and Biosynthesis. *Chem. Rev.* **2005**, *105* (2), 621−632.

(4) Fukuyo, Y.; Hunt, C. R.; Horikoshi, N. Geldanamycin and Its Anti-Cancer Activities. *Cancer Lett.* **2010**, *290* (1), 24−35.

(5) Valdes-Pena, M. A.; Massaro, N. P.; Lin, Y.-C.; Pierce, J. G. Leveraging Marine Natural Products as a Platform to Tackle Bacterial Resistance and Persistence. *Acc. Chem. Res.* **2021**, *54* (8), 1866−1877.

(6) Hager, A.; Kuttruff, C. A.; Herrero-Gomez, E.; Trauner, D. Toward the Total Synthesis of Ansalactam A. *Tetrahedron Lett.* **2014**, *55*, 59−62.

(7) Breugst, M.; Reissig, H. The Huisgen Reaction: Milestones of the 1,3-Dipolar Cycloaddition. *Angew. Chem., Int. Ed.* **2020**, *59* (30), 12293−12307.

(8) Narayan, R.; Potowski, M.; Jia, Z.-J.; Antonchick, A. P.; Waldmann, H. Catalytic Enantioselective 1,3-Dipolar Cycloadditions of Azomethine Ylides for Biology-Oriented Synthesis. *Acc. Chem. Res.* **2014**, *47* (4), 1296−1310.

(9) Kaur, T.; Wadhwa, P.; Bagchi, S.; Sharma, A. Isocyanide Based [4 + 1] Cycloaddition Reactions: An Indispensable Tool in Multi-Component Reactions (MCRs). *Chem. Commun.* **2016**, *52* (43), 6958−6976.

(10) Kumar, P. Synthesis of substituted 1-tetralones. *Org. Prep. Proced. Int.* **1997**, *29*, 477−480.

(11) Kurata, H.; Kusumi, K.; Otsuki, K.; Suzuki, R.; Kurono, M.; Komiya, T.; Hagiya, H.; Mizuno, H.; Shioya, H.; Ono, T.; Takada, Y.; Maeda, T.; Matsunaga, N.; Kondo, T.; Tominaga, S.; Nunoya, K.; Kiyoshi, H.; Komeno, M.; Nakade, S.; Habashita, H. Discovery of a 1-Methyl-3,4-dihydronaphthalene-Based Sphingosine-1-Phosphate (S1P) Receptor Agonist Ceralifimod (ONO-4641). A $S1P_1$ and $S1P_5$ Selective Agonist for the Treatment of Autoimmune Diseases. *J. Med. Chem.* **2017**, *60*, 9508−9530.

(12) Mohamed, M.; Yamada, K.; Tomioka, K. Accessing the amide functionality by the mild and low-cost oxidation of imine. *Tetrahedron Lett.* **2009**, *50*, 3436−3438.